

Tarun Joseph

Machine Learning Engineer | AI Program Design

Location: Toronto, Canada

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TECHNICAL SKILLS

Programming: Python, R, Java, SQL, Git | **AI & LLM Tools:** OpenAI APIs, Claude Code, GitHub Copilot, Codex, LangChain | **Frameworks & Cloud:** TensorFlow, PyTorch, GCP, AWS, Azure

PROFESSIONAL EXPERIENCE

- ❖ **Applied AI Project Management Intern | Vector Institute** May 2026 – Present
 - **Program Design:** Designing and delivering Applied AI programs that bridge cutting-edge AI research and practical industry implementation across Canada's AI ecosystem.
 - **Stakeholder Management:** Serve as liaison between Vector Institute researchers and industry sponsors, translating technical constraints into clear project roadmaps and deliverable timelines.
 - **Technical Translation:** Facilitate cross-functional workflows by communicating AI requirements and deployment considerations to non-technical partners, ensuring alignment from scoping through delivery.
- ❖ **Machine Learning Engineer | Tata Consultancy Services (TCS)** January 2023 – May 2025
 - **AI Prototyping:** Deployed GPT-3.5 and Llama 3-powered RAG chatbots and AI agents for TCS' Edtech platform, benefiting 600,000 associates globally.
 - **Workflow Automation:** Designed a PoC and implemented an MVP for support ticket automation using open-source LLMs, successfully eliminating redundant manual efforts and saving over five hours per developer per week.
 - **Client Advisory & Teaching:** Led deep-skilling Generative AI bootcamps for 10,000+ associates, teaching prompt engineering and LLM application development, which drove a 13% increase in platform engagement.
 - **Optimization:** Automated redundant manual tasks across developer teams using Generative AI tools, reducing overall manual effort by 30%.
- ❖ **Data Analyst | Tata Consultancy Services (TCS)** July 2021 – December 2022
 - **Data Integration:** Developed a unified SQL database and automated recommendation mailers using Python, reducing manual administrative effort by 50%.
 - **Metrics & Reporting:** Analyzed platform data to identify learning drop-offs, implementing insights that successfully boosted program engagement by 11%.

TEACHING & CAMPUS LEADERSHIP

- ❖ **Graduate Teaching Assistant | University of Toronto** September 2025 – April 2026
 - **MIE429 AI/ML Capstone Design:** Mentored 45 final-year students in designing AI prototypes, stepping in to translate technical roadblocks during meetings with industry clients to ensure successful project delivery.
 - **APS106 (Fundamentals of Computer Programming) & MIE1628 (Cloud-based Big Data Analytics):** Supervised labs for 500+ students, teaching Python algorithms, debugging code, and managing high volumes of student inquiries to support their technical transitions.
- ❖ **Team Lead | Deep Learning Graduate Project | University of Toronto** January 2026 – April 2026
 - **Project Management:** Led a four-person team to build a real-time American Sign Language (ASL) translator using computer vision with 65% reliability and secured A+ grade for the team.
- ❖ **VP of Professional Development | MIE Graduate Student Association (AMIGAS)** January 2026 – Present
 - **Workshop Facilitation:** Design educational resources and lead professional development workshops to clearly explain complex topics to graduate students.
- ❖ **Departmental Steward | CUPE 3902 Unit 1 | University of Toronto** January 2026 – Present
 - **Stakeholder Liaison & Communication:** Act as a bridge between administration and student workers by translating complex policy updates into clear, actionable guidelines and resolving any grievances.

EDUCATION & CERTIFICATIONS

- ❖ **MEng, Industrial Engineering | University of Toronto, Canada** September 2025 – Present
 - **Relevant Courses:** Deep Learning, Data Science & Statistics, HR Management, Management Consulting
 - **Emphasis:** Data Analytics & Machine Learning, ELITE | GPA: 3.8/4
- ❖ **B. Tech, Industrial Engineering | College of Engineering Trivandrum, India** August 2017 – June 2021
- ❖ **Certifications:** Google: Generative AI Leader Certification (2026) | AWS Certified AI Practitioner (2024) | Microsoft Certified Azure Data Scientist Associate (2024) | AWS Certified Cloud Practitioner (2023)